Similarly, $15\sin(\theta + 60^\circ) = \frac{3}{2}(4 + 3\sqrt{3})$. Hence, the point B is given by

$$B\left(\frac{3}{2}(3 - 4\sqrt{3}), \frac{3}{2}(4 + 3\sqrt{3})\right).$$

Example 3.19 A ripple tank demonstrates the effect of two water waves being added together. The two waves are described by $h = 8\cos t$ and $h = 6\sin t$, where $t \in [0, 2\pi)$ is in seconds and h is the height in millimeters above still water. Find the maximum height of the resultant wave and the value of t at which it occurs.

Solution:
Let H be the height of the resultant wave at time t. Then H is given by

$$H = 8\cos t + 6\sin t$$

Let $8\cos t + 6\sin t = k\cos(t - \alpha) = k(\cos t\cos\alpha + \sin t\sin\alpha)$

Hence, $k = 10$ and $\tan\alpha = \frac{3}{4}$, so that

$H = 10\cos(t - \alpha)$

Thus, the maximum of $H = 10$ mm. The maximum occurs when $t = \alpha$, where $\tan\alpha = \frac{3}{4}$.

Example 3.20 Expand (i) $\sin(A + B + C)$ (ii) $\tan(A + B + C)$

Solution:

$$\begin{aligned}
\text{(i) } \sin(A + B + C) &= \sin[A + (B + C)] \\
&= \sin A\cos(B + C) + \cos A\sin(B + C) \\
&= \sin A\cos B\cos C + \cos A\sin B\cos C \\
&\quad + \cos A\cos B\sin C - \sin A\sin B\sin C \\
\text{(ii) } \tan(A + B + C) &= \tan[A + (B + C)] \\
&= \frac{\tan A + \tan(B + C)}{1 - \tan A\tan(B + C)} \\
&= \frac{\tan A + \dfrac{\tan B + \tan C}{1 - \tan B\tan C}}{1 - \tan A\dfrac{\tan B + \tan C}{1 - \tan B\tan C}} \\
&= \frac{\tan A + \tan B + \tan C - \tan A\tan B\tan C}{1 - \tan A\tan B - \tan B\tan C - \tan C\tan A}
\end{aligned}$$

(i) If A+B+C = 0 or π, we have $\tan(A + B + C) = 0$ so that

$$\tan A + \tan B + \tan C = \tan A\tan B\tan C.$$

This result is also true in the case of oblique triangles.

(ii) $\tan(x - y) + \tan(y - z) + \tan(z - x) = \tan(x - y)\tan(y - z)\tan(z - x)$